

CENTRAL PHILIPPINE UNIVERSITY

Running Case2: QuickBiz

Members:

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1.1 Case Summary

Andrew Langston founded QuickBiz Messenger, a package delivery service company. Langston used to transport packages on a bicycle with no communication in the beginning except for the pager, medium. The company's messengers had major communication breakdowns. However, as new technology such as cell phones and email became available.

The corporation kept electronic personnel files concerning each employee's performance to assure quality. Andrew Langston wrote all of the log sheets. He had no computers. Andrew employed Sarah as his business grew. Truesdale will be his secretary and bookkeeper. Sarah set up the office and organized it.

On the company's first PC, word processing, spreadsheet, and database tools were installed.

1.2 Key Issue in the Case

1. Courier and Customers Data Information

The company uses an unmaintained database for their employees and couriers while the technology is advancing.

The company were using word processing and spreadsheet for them to keep the information. It is impossible to run and manage data effectively without a database management system. A DBMS, which acts as an intermediary between the user and the database, gives users access

to files stored in a database. It gives the end user a unified view of the data and converts all applications it receives into complex operations that fulfill those requests.

1.3 Recommendations

We've come up with a few recommendations based on the difficulties listed above.

To solve these, use QuickBiz Messengers:

1.Update and improve the Database system.

QuickBiz Messenger needs a Database that would be efficient and convenient for both of the staffs and the customers. Companies need to keep basic transaction records. Provide information that will assist the company in running its operations more efficiently.Database systems are critical to your business because they communicate data about your sales transactions, product inventory, customer profiles, and marketing activities.

2. User Friendly App/System.

For we are having a system for the company to use we also need a user-friendly interface and simple navigation. It the user reduces search time and increases satisfaction, allowing him to meet his needs quickly and efficiently. As a result, the brand increases sales volume, increases

customer loyalty, and reduces costs and resources. Both the client and the company are interested in knowing whether the goods were delivered on time.

1.4 Scope and Limitation:

Scope

The scope of the study is about technological problems. The owner wants to record the data easily, like recording the data of the deliveries, the employees' data, and how all the things will be delivered on time. Since technology is upgrading every time, it can help with the problem of Andrew Langston. This will also be easy, less hassle and an advantage for the customers and also for the employees. Like the employees' data, if their data is accidentally lost, you can just retrieve it back easily because of upgraded technologies. And also about deliveries: it can help the company have good feedback because they deliver the customers' packages or things on time. This may help with the solutions for the problems, like using digital format and tracking devices.

Limitations

This study serves as information about making Andrew's problem solved. Because recording data is hard by just writing it on paper or anything, it will be lost easily. That is why we have technology. And also the risks that they take just to solve the problem to help the company, like how hard it is for the bikers to deliver packages or things. That it can disturb the flow of business, and as result make unhappy customers. For instance, he should purchase a set of

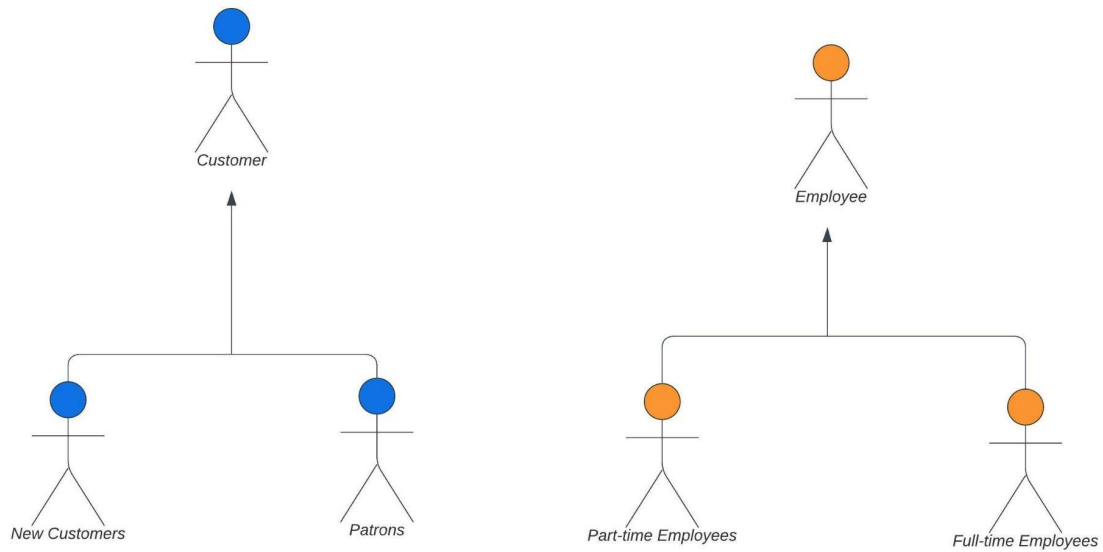
computers and improvise navigation through deliveries. Therefore, the limitation of this study is that it has not been proven to be effective or to cause a positive change.

Significance of the study

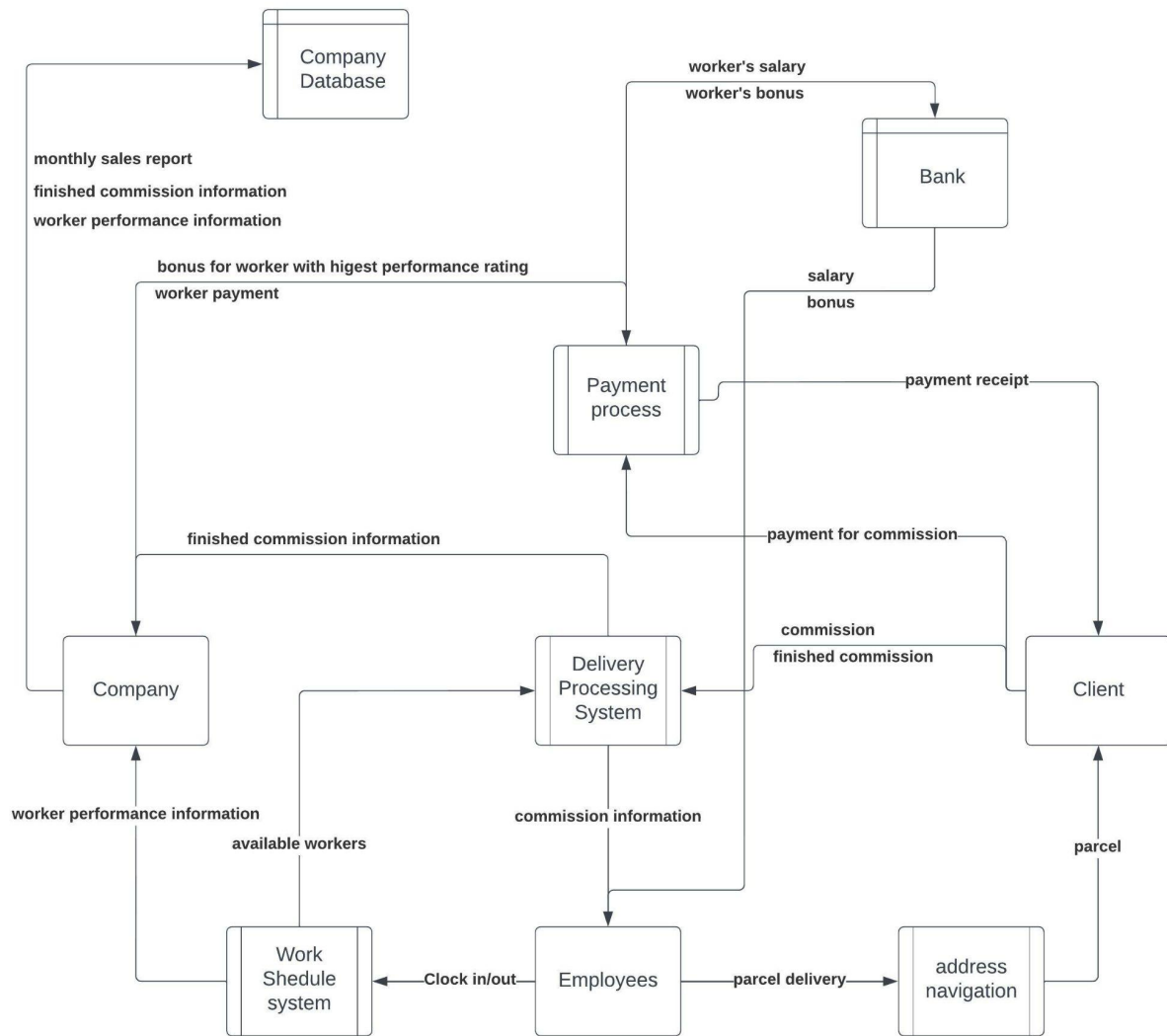
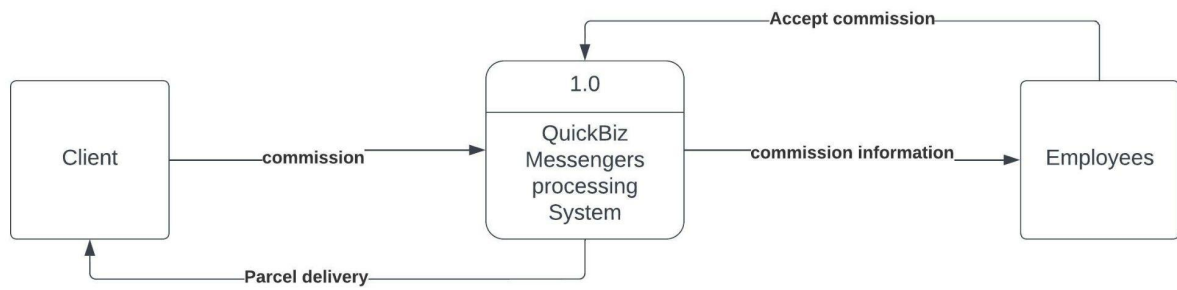
The new system will make commission creation easier for customers. The company's commission list will also be made more accessible and easy to track for the company's employees, making deliveries faster and easier for their couriers.

Chapter 2

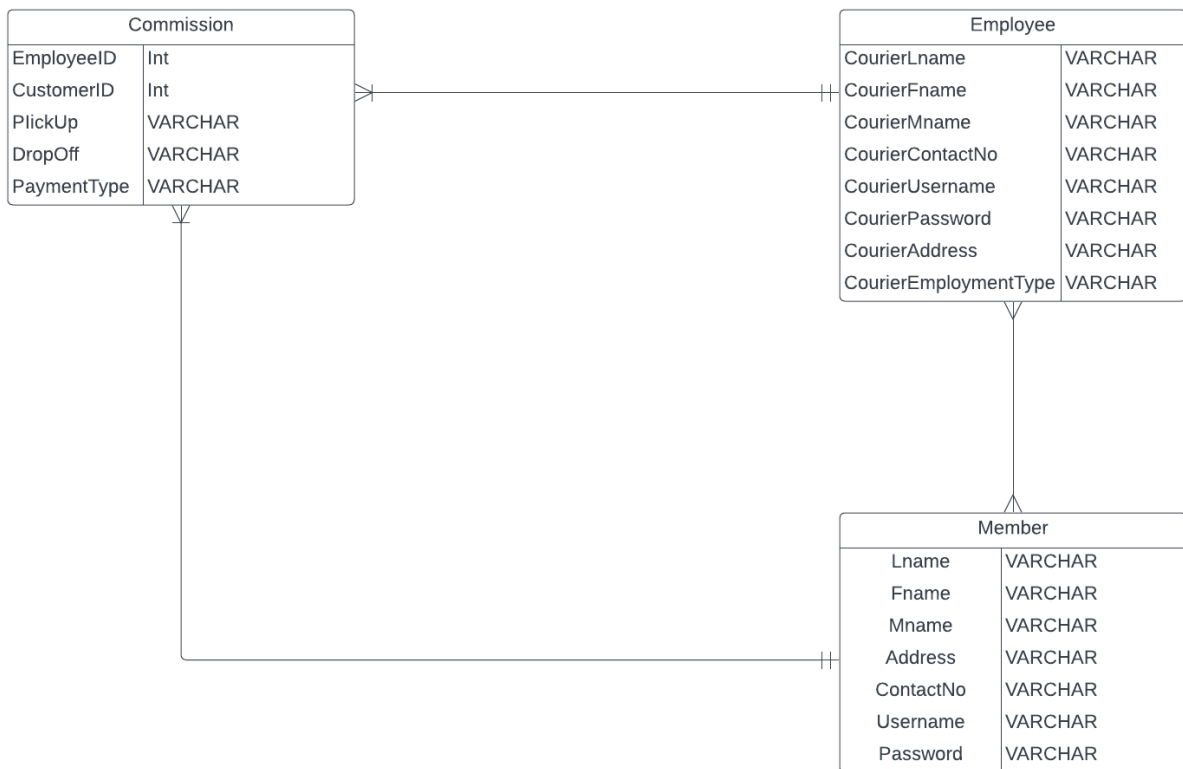
2.1 Users of the system



2.2 Proposed Data Flow Diagram



2.3 Entity Relationship Diagram



2.4 Database Structure

Table Name: Commission

Primary Key: ID

Foreign Key: Employee ID

Customer ID

Table Description: This table is used to store commissions data.

Field	Attributes	Data Type	Size	Content/Description
1	id	Integer	20	ID
2	employeeID	Integer	20	Employee ID
3	customerID	Integer	20	Customer ID
4	pickup	Varchar	50	Pick Up
5	dropoff	Varchar	50	Drop Off
6	paymenttype	Varchar	50	Payment Type

Table Name: Employee

Primary Key: ID

Foreign Key: NA

Table Description: This table is used to store employees data.

Field	Attributes	Data Type	Size	Content/Description
1	id	Integer	20	ID
2	courierLname	Varchar	50	Courier Last Name
3	courierFname	Varchar	50	Courier First Name
4	courierMname	Varchar	50	Courier Middle Name

5	courierContactNo	Varchar	20	Courier Contact Number
6	courierAddress	Varchar	150	Courier Address
7	courierUsername	Varchar	50	Courier Username
8	courierPassword	Varchar	50	Courier Password
9	courierEmploymenttype	Varchar	50	Courier Employment Type

Table Name: Member

Primary Key: ID

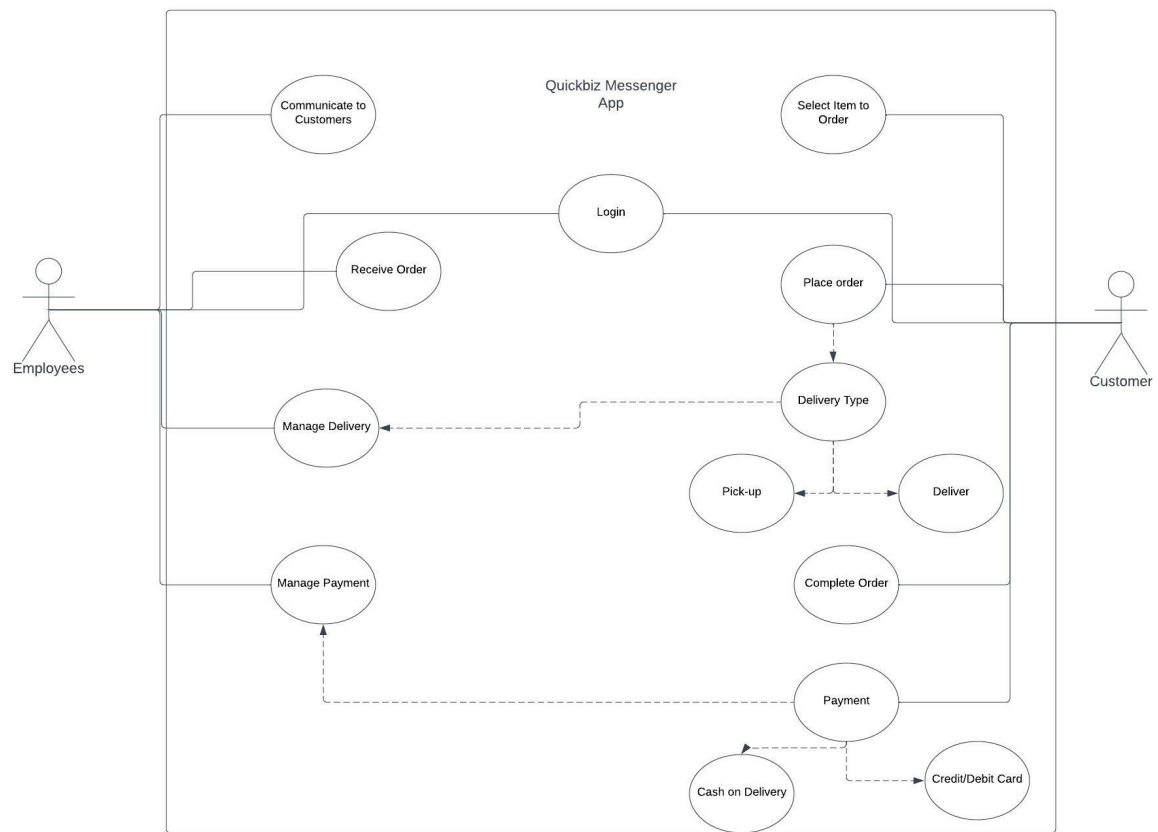
Foreign Key: NA

Table Description: This table is used to store members data.

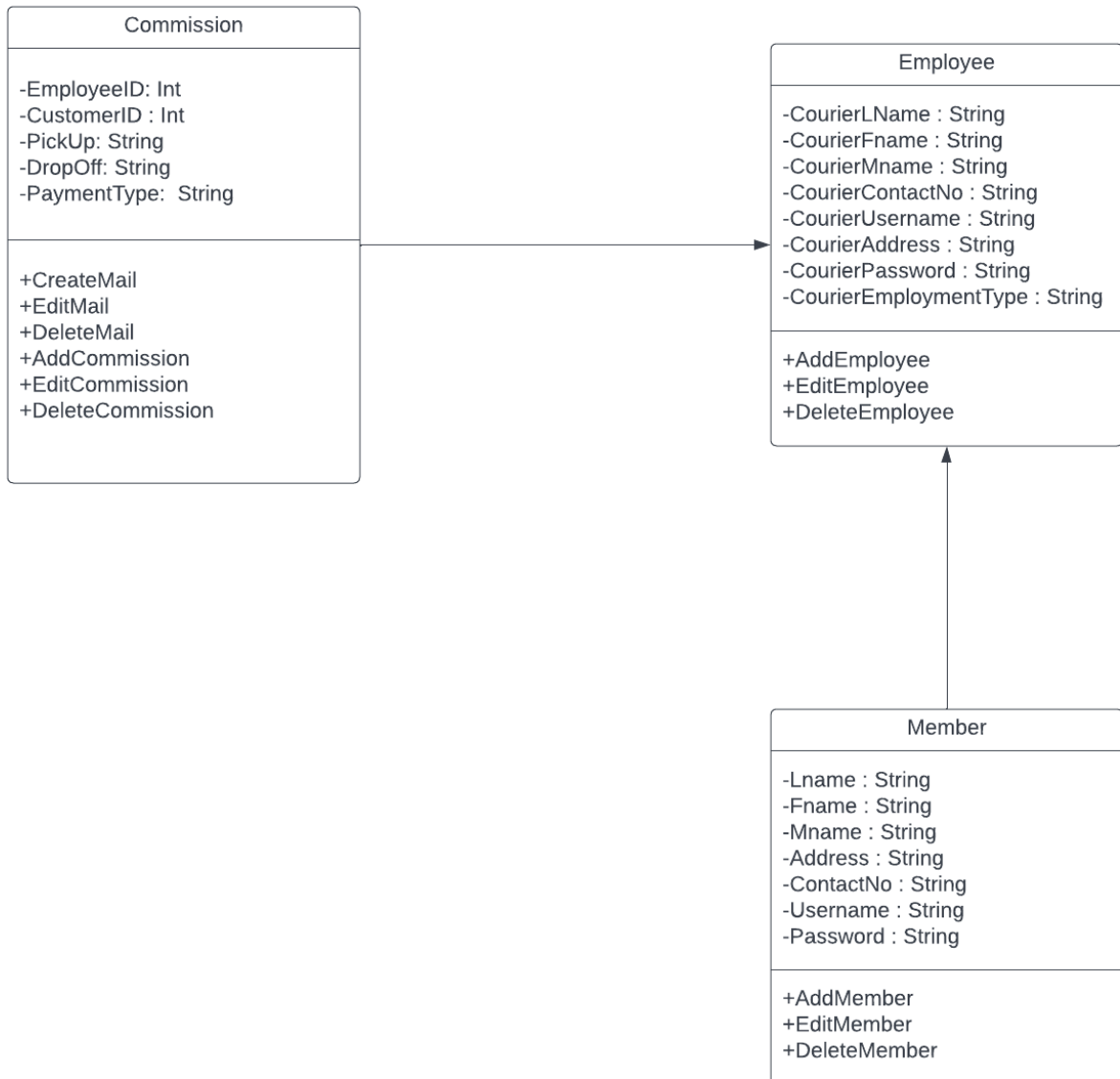
Field	Attributes	Data Type	Size	Content/Description
1	id	Integer	20	ID
2	Lname	Varchar	50	Last Name
3	Fname	Varchar	50	First Name
4	Mname	Varchar	50	Middle Name
5	address	Varchar	50	Address

6	contactNo	Varchar	50	Contact Number
7	username	Varchar	50	Username
8	password	Varchar	50	Password

2.5 Use-case Diagram



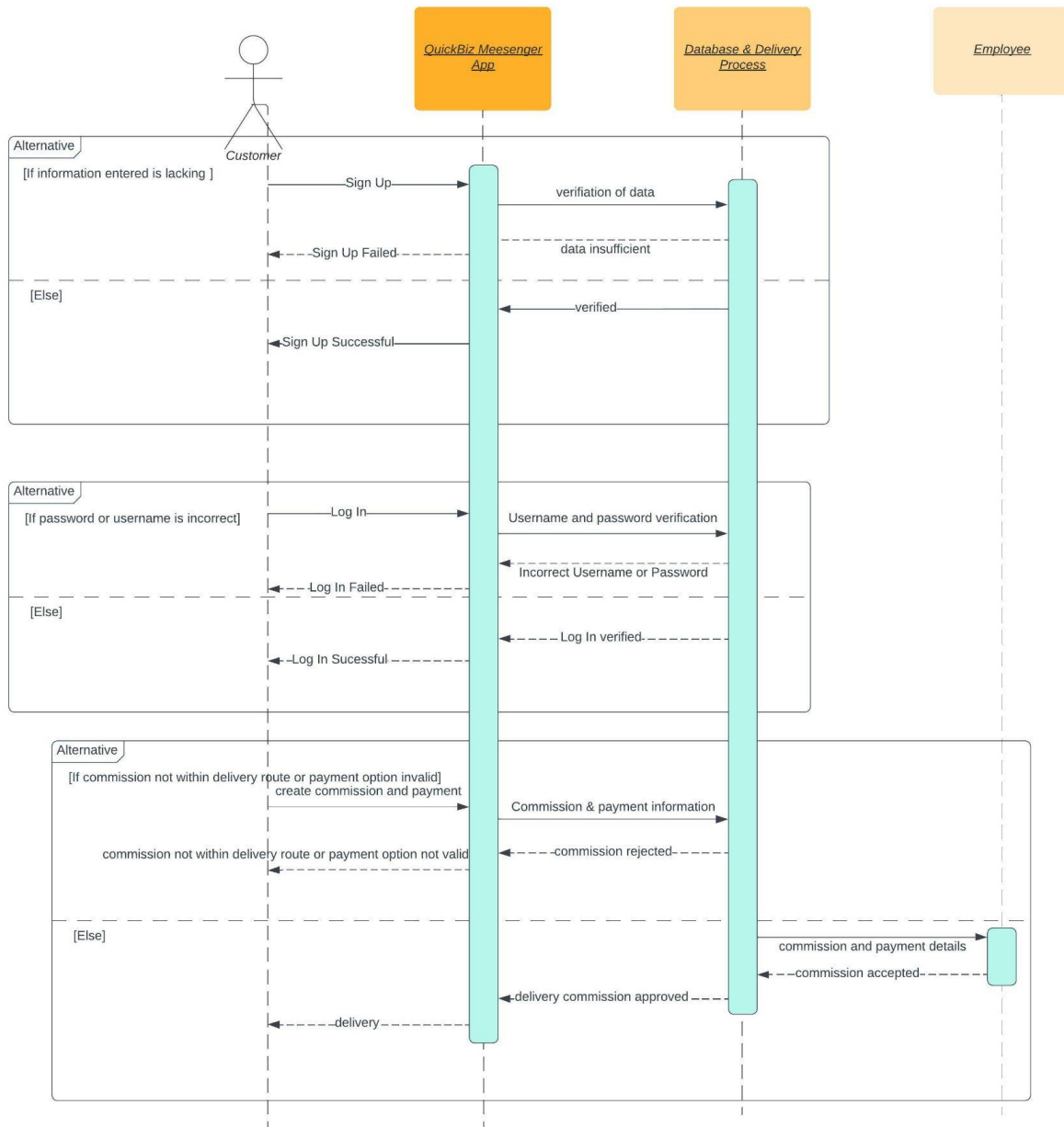
2.6 Class Diagram



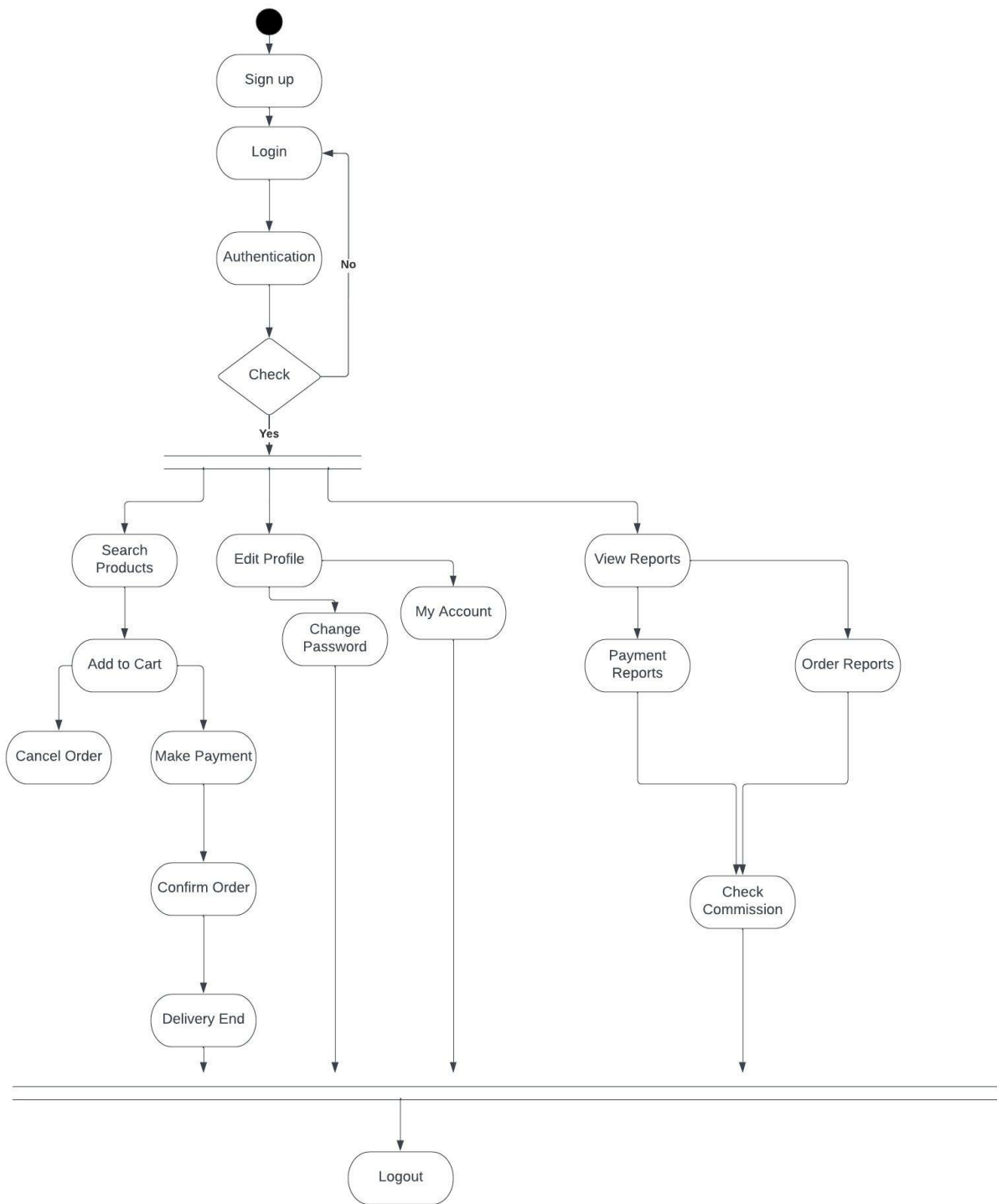
2.7 Sequence Diagram

Sequence diagram Case 2

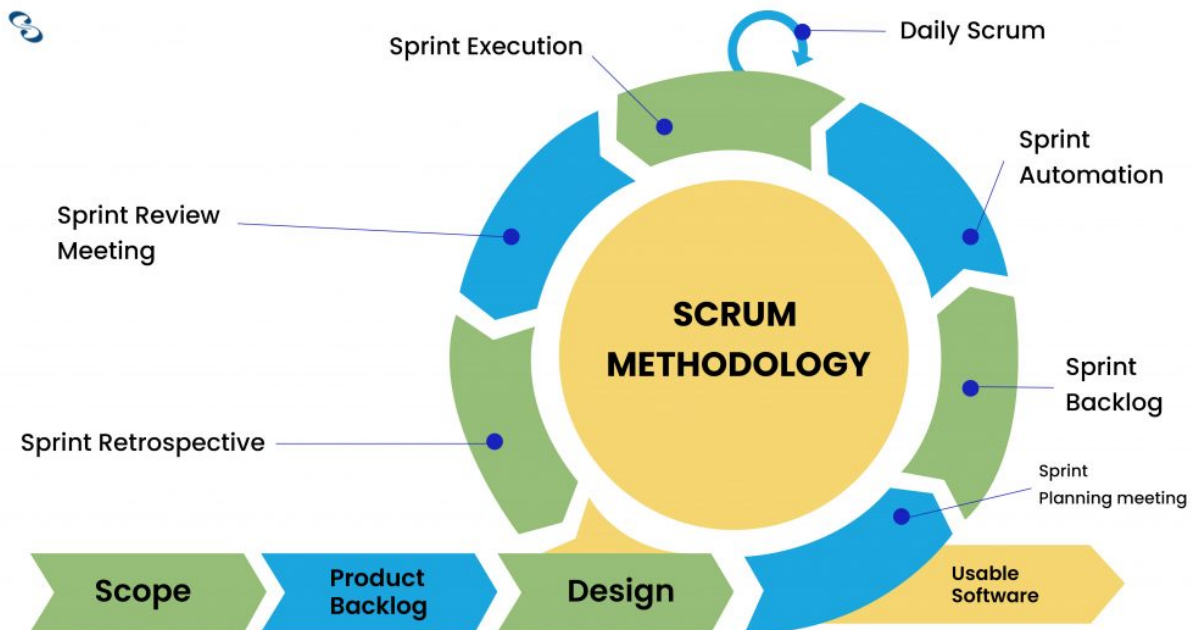
Norberto O. BABIERA III | June 18, 2022



2.8 Activity Diagram



Chapter 3: Methodology



SCRUM is a methodology for creating, delivering, and maintaining products in a complex environment. It was originally developed for software development, but it has now been used in various industries such as research, sales, marketing, and advanced technologies.

We have chosen to use this methodology because it keeps our priorities straight, short-term plans make it easier to perform tasks, and it helps us organize our schedule while executing our daily tasks by planning ahead of time.

Vision

The researchers conducted a preliminary investigation by gathering information on the client's present system and what he envisions the project will include. After creating a plan, the researchers test its feasibility, checking for problems and possible improvements to the initial plan.

Sprint Planning

During this stage, after a thorough investigation, the researchers identified the problems to be addressed and improvements to be implemented by the new system. The researchers create a product item list, including the new features the new system will implement. The researchers then create a sprint tasklist regarding the tasks that need to be done to finish the items in the product item list. Each task in the Sprint backlog corresponds to the product backlog.

Sprint 1

In the first sprint, the researchers gathered data, reviewed the client's vision for the new system, and conducted a thorough analysis of the new system. The researchers then created multiple DFDs to give shape to the new system's structure. The researchers then included the manager in the evaluation of the new system proposed. After receiving approval, the researchers began to plan for the next sprint listed in the sprint backlog.

Sprint 2

In the second sprint, the researchers created an initial design for the system, which was coded by the developer. After creating the login, sign up, commission, worker database, and commission list modules, the developer coded the module and started testing the code for errors and bugs, repeating the evaluation process until the code was finally finished.

Sprint 3

In the third sprint, after the developer finished the project, the researchers reevaluated their program with the manager to make sure that all requirements were met. After re-evaluation, more testing has been done.

Sprint Review

After all the tasks had been completed, the researchers reviewed the sprint backlog to ensure all tasks had been completed and requirements have all been met.

Chapter 4: Budget Proposal

I. Hardware Specification

Input Devices	LOGITECH MK295 SILENT WIRELESS KEYBOARD AND MOUSE COMBO	₱1,550.00
Operating System	Windows 10 Pro	₱10,752.00
Output Devices	SAMSUNG 23.5" CF396 Curved Computer Monitor	₱8,601.00
Storage Devices	SAMSUNG 970 EVO Plus SSD 2TB	₱10,429.00
Memory	G.SKILL TRIDENT Z RGB 16GB (2X8GB) DDR4-3600MHZ F4-3600C18D-16GTZR Memory	₱6,850.00
CPU	Intel Core i5-10600 Processor (12M Cache, 6 Cores, 12 Threads, up to 4.80 GHz)	₱13,950.00
UPS	APCEASY UPS BV 1000VA, AVR, Universal Outlet, 230V	₱5,950.00
TOTAL:		₱57,782.00

II. Software Specification

Operating System	Windows 10 Pro
Back End	MySQL, <u>phpMyAdmin</u>
Front End	C#, Visual Studio 2019

III. Mobile user Specification

Operating System	Android 6 or higher
Networks	GSM,LTE,HSPA
Storage	16GB or higher
Memory	2GB

IV. Soft Development Cost

Software Development Team	
Project Manager	₱41,000.00
Software Developer	₱36,200.00
Analyst	₱24,500.00
Consultant	₱27,800.00
Total:	₱129,500.00

V. Operating System Choice

Microsoft Windows 10 Pro was chosen as the minimum requirement as it is an operating system still supported and updated by Microsoft and has the minimum level of security against modern security threats.

VI.Front End

-C#

C# was the programming language chosen because it is a Microsoft object-oriented programming language that tries to combine C++'s computing capability with Visual Basic's programming ease. C# is based on C++ and has features that are similar to Java, allowing it to create a broad range of programs, including Web and Mobile applications.

-Visual Studio 2019

is an IDE (integrated development environment) software developed by Microsoft used for different types of software development such as computer programs, websites, web apps, web services, and mobile apps. It contains completion tools, compilers, and other features to facilitate the software development process.

Microsoft Visual Studio was chosen because of its ease of use when creating web apps with .NET or ASP.NET, making coding easier and more efficient.

VII.Depreciation

Software depreciation

Software Depreciation Rate (Rate of Depreciation = Cost/Year)

Software Cost = Php 129,500

Deprecation Year = 10 Years

Rate of Deprecation = Php 11,950.00

Year	Amount
0	₱129,500
1	₱117,550
2	₱105,600
3	₱93,650

4	₱81,700
5	₱69,750
6	₱57,800
7	₱45,850
8	₱33,900
9	₱21,950
10	₱0

Hardware depreciation

Hardware Depreciation Rate (Rate of Depreciation = Cost/Year)

Hardware Cost = Php 57,782

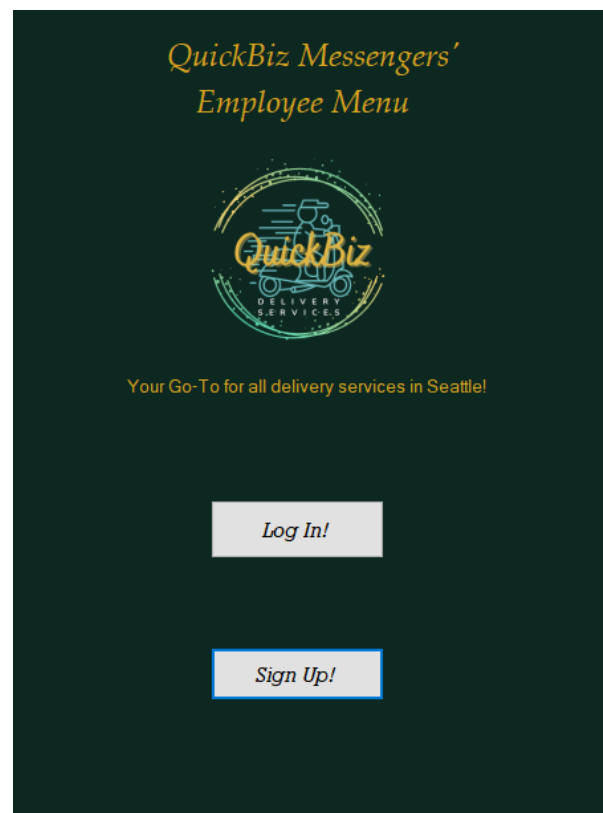
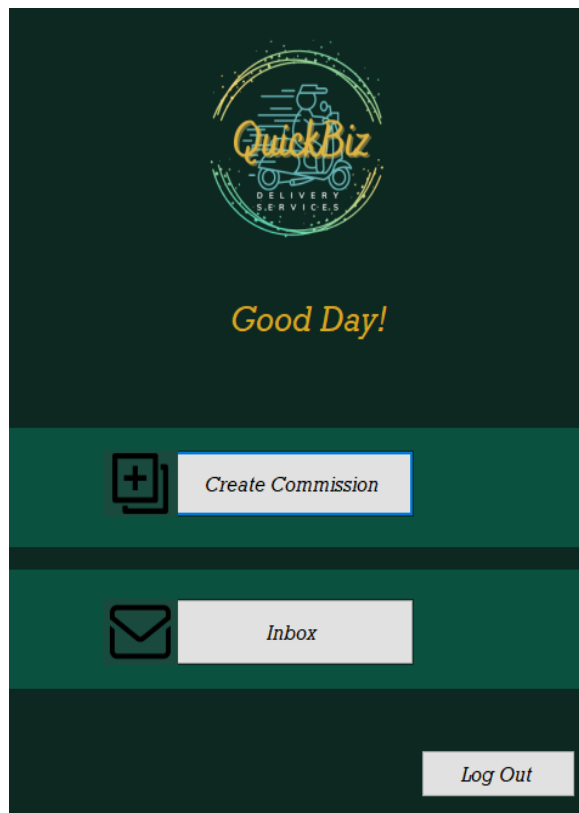
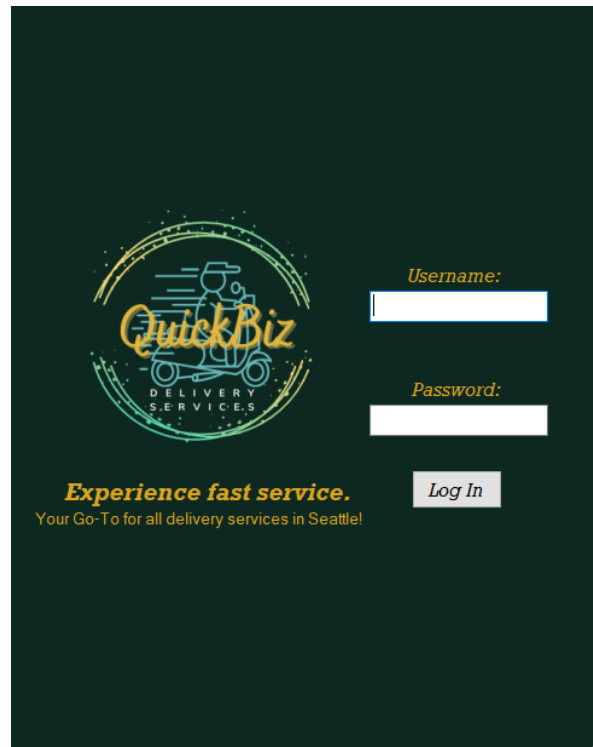
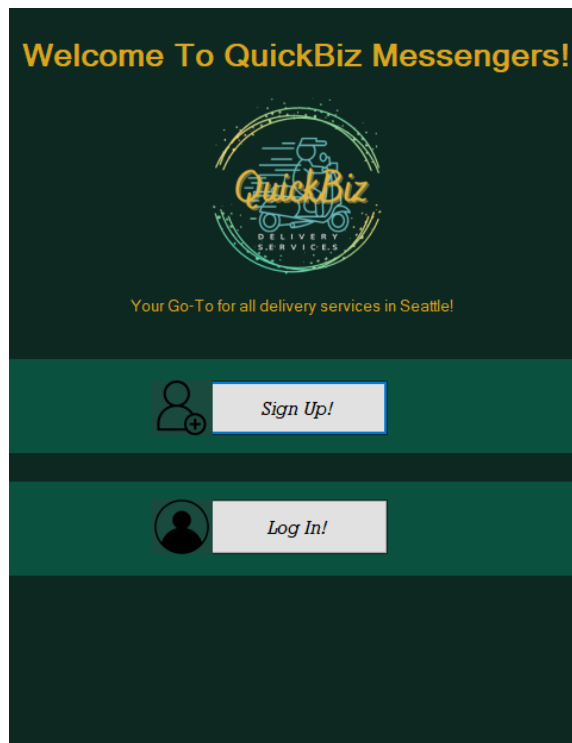
Deprecation Year = 5 Years

Rate of Deprecation = Php 11,556.4

0	₱57,782
1	₱46,225.6
2	₱34,669.2
3	₱23,112.8
4	₱11,556.4
5	₱0

VIII. Appendices

Screenshots



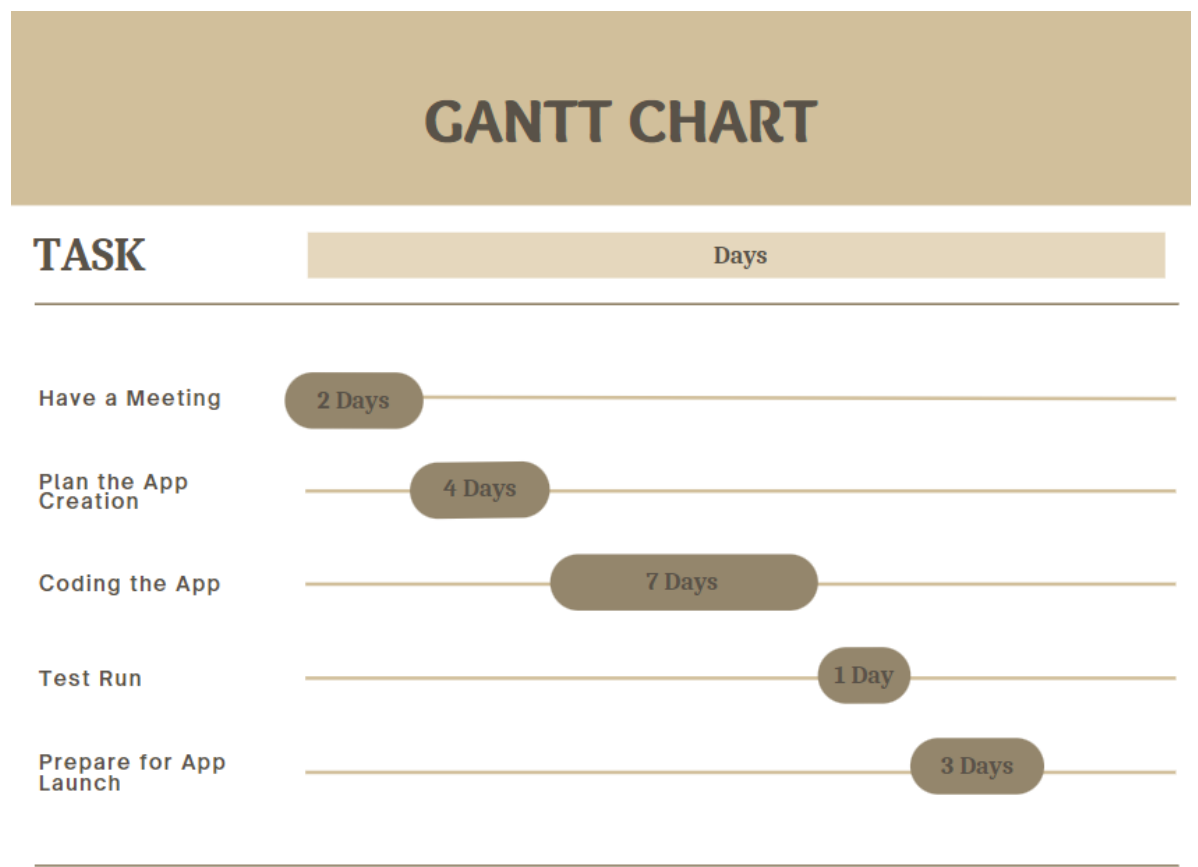


Commissions

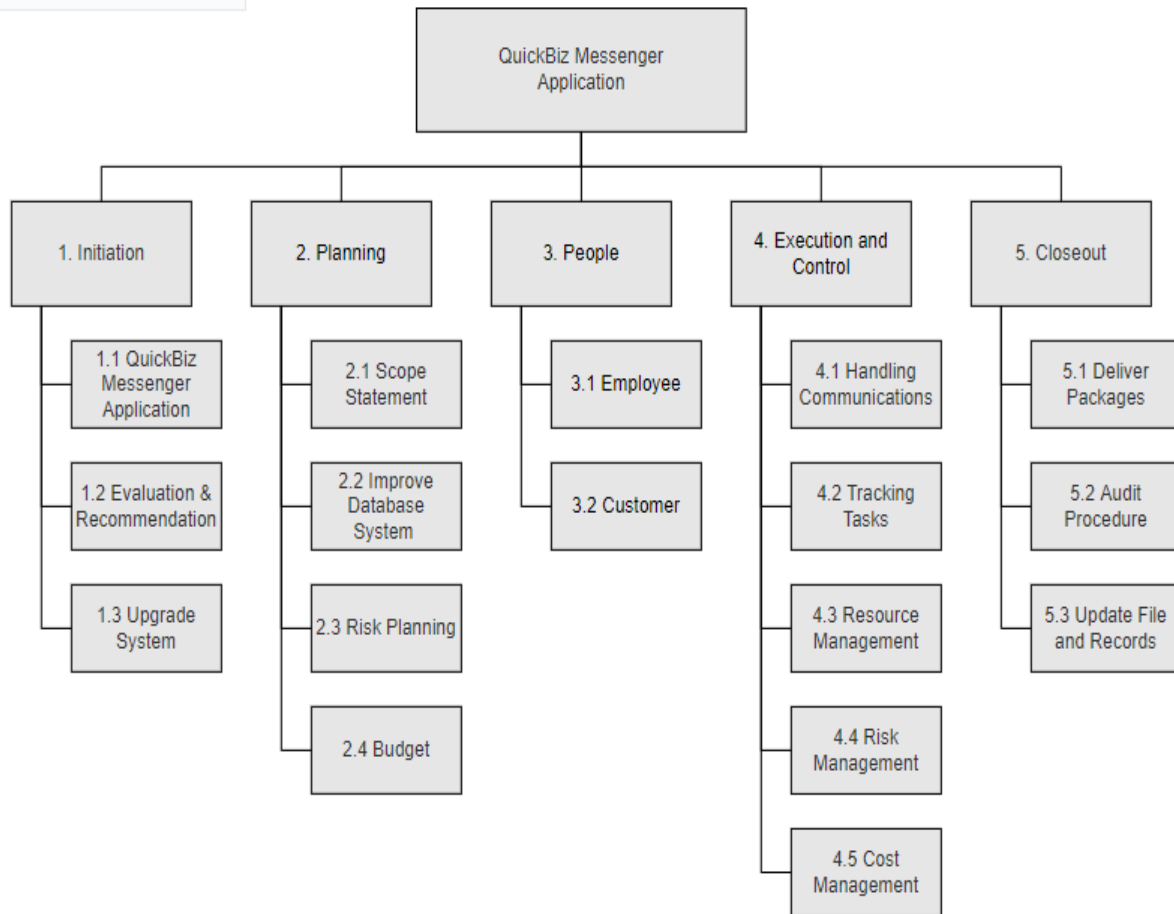
	ID	Pick Up Address	Drop Off Address	Payment Type	CustomerID	CourierID

return to dashboard

Gannt Chart



Work Breakdown Structure



DFD

